

Cryptanalysis of a Medical Image Encryption Scheme Using Multiple Chaotic Maps

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Abstract

A chaotic image encryption scheme combining Arnold's Cat Map with a Two-Dimensional Logistic Sine Coupling Map was proposed by Jain et al [1] . for securing medical images. The scheme follows a permutation-diffusion architecture and was reported to achieve satisfactory resistance against statistical and differential attacks. In this paper, a detailed cryptanalysis is presented, demonstrating that the scheme cannot resist a chosen-plaintext attack due to inherent structural weaknesses.

For the chosen-plaintext attack, first, an image with all pixel values equal to zero is selected, and its corresponding ciphertext is obtained. Through simple modular arithmetic analysis of the diffusion process, the complete equivalent keystream is recovered. Second, a set of specially constructed probe images is chosen to determine the column-wise permutation mapping, where the previously recovered keystream is used to eliminate the diffusion effect from the ciphertexts. Finally, by combining the recovered keystream and permutation mapping, any target ciphertext can be decrypted exactly without knowledge of the secret key.

Theoretical analysis and experimental results demonstrate that the attack requires only a small number of chosen plaintexts proportional to the image dimension and achieves lossless reconstruction. The analysis in this

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work provides useful insight for improving the design of chaos-based image encryption algorithms.

Keywords: Chaotic image encryption, Cryptanalysis, Chosen-plaintext attack, Keystream recovery, Permutation reconstruction, Arnold’s Cat Map, 2D-LSCM

1. Introduction

Advances in digital networking have made image-based data exchange central to many critical domains, including telemedicine, video surveillance, and multimedia content delivery [2]. The widespread adoption of cloud platforms, Internet-of-Things devices, and mobile communication infrastructure has driven an explosive growth in the volume of images produced and transmitted globally [3]. However, the inherently open character of contemporary communication networks creates serious vulnerabilities: images are susceptible to interception, unauthorized disclosure, and deliberate tampering [4, 5]. Applications such as remote clinical diagnostics and biometric identification are especially exposed, since a single breach can compromise both individual privacy and institutional operations [6]. Consequently, robust protection of digital imagery has become an indispensable component of modern information security infrastructure [7].

Numerous image-specific encryption approaches have been investigated as countermeasures to these threats. Unlike generic data, digital images are characterized by substantial pixel redundancy, pronounced spatial correlations among neighboring pixels, and large file sizes [8]. Conventional block ciphers such as AES and RSA, while secure for arbitrary data, are ill-suited to these characteristics and impose prohibitive computational overheads when applied directly to high-resolution images [9]. To address this mismatch, researchers have developed dedicated image encryption schemes leveraging chaos theory, bitwise logic, and various mathematical transforms [10].

Chaos-based image encryption has attracted considerable interest owing to the natural alignment between the properties of chaotic dynamical systems—sensitivity to initial conditions, ergodicity, and topological mixing—and the desiderata of sound image encryption, namely confusion and diffusion [11]. More recently, chaotic approaches have been augmented with DNA coding [12], compressed sensing [13], bit-level operations [14], and frequency-domain transforms [15] to extend their security margins and broaden

applicability.

A landmark contribution to the field was made by Fridrich [16], who formalized the permutation-diffusion architecture based on Baker’s map—a two-stage framework that has since become the de facto design template for chaos-based image ciphers. In this architecture, the permutation stage scrambles pixel locations to break spatial redundancy, while the diffusion stage propagates the influence of each plaintext pixel throughout the entire ciphertext.

Within this framework, Guan et al. [17] devised an encryption scheme coupling Arnold’s cat map with Chen’s three-dimensional chaotic attractor, targeting high confusion levels. Wu et al. [18] explored a two-dimensional logistic map for both the permutation and substitution stages, providing theoretical analysis of the resulting confusion and diffusion properties. Mursi et al. [19] further enriched this line of work by incorporating the fractional Fourier transform and a modified Hénon map, achieving broader parameter ranges and improved statistical behavior. Contemporary efforts have shifted attention toward constructing novel chaotic systems specifically engineered to overcome the limitations such as small chaotic windows and periodic windows—inherent in lower-dimensional maps.

The Logistic Sine Cosine Image Encryption Scheme (LSC-IES) [1] addresses initialization sensitivity by combining logistic and sine maps with a cosine transform, iterating the joint system multiple times to maximize entropy in the output. Thumbnail-preserving chaos-based variants [20] and dedicated color-image ciphers [21] illustrate how the permutation-diffusion paradigm continues to evolve in response to diverse application requirements.

Despite steady advances in the design of chaotic ciphers, a parallel body of work has systematically revealed cryptographic weaknesses arising from poor coupling between the keystream and the plaintext [22]. Xie et al. [23] demonstrated that Fridrich’s original Baker-map cipher leaks plaintext information into the ciphertext in a recoverable manner. Subsequently, Cokal and Solak [24] established that the Arnold-and-Chen scheme of Guan et al. succumbs to both known-plaintext and chosen-plaintext attacks, inaugurating a pattern in which statistically strong ciphers collapse under algebraic cryptanalysis.

The fundamental vulnerability in most chaotic ciphers is that their pseudorandom keystreams are generated solely from the secret key, independently of the plaintext. A keystream that does not vary with the plaintext is functionally equivalent to a reusable one-time pad and is therefore

trivially vulnerable to chosen-plaintext attacks [25, 26]. In the VHC-CIES scheme [27], for example, the key tuple $(x_0, y_0, z_0, \mu, r_1, r_2)$ produces pseudo-random sequences without any plaintext involvement, rendering the scheme fully breakable through algebraic analysis of the Hill cipher variant and subsequent keystream extraction [28, 29]. Analogous flaws oversimplified linear transformations and static equivalent keys have been uncovered in numerous other recently proposed schemes [30].

The accumulated evidence from this body of cryptanalytic literature establishes a clear principle: satisfactory performance on statistical benchmarks does not constitute cryptographic security, and formal cryptanalysis is an obligatory step before any cipher can be considered suitable for deployment [31].

With this perspective in mind, the present paper re-examines the scheme of Jain et al. [1], which employs Arnold’s Cat Map for pixel shuffling and a 2D Logistic Sine Coupling Map for diffusion within a two-round permutation-diffusion structure. The authors of the original scheme claimed strong resistance to both statistical and differential attacks for medical imaging applications. We demonstrate that these claims do not extend to chosen-plaintext scenarios: the plaintext-independent and deterministic character of both the permutation and diffusion layers makes it possible to recover the equivalent keystream and the complete permutation map using a modest number of oracle queries, thereby enabling decryption of any ciphertext without the secret key.

The contributions of this paper are as follows. A rigorous theoretical analysis identifies the structural root causes of the vulnerability. An efficient three-phase CPA is constructed, recovering the keystream in a single oracle query and the full permutation map in n additional queries for an $n \times n$ image. Experimental validation on a standard medical test image confirms pixel-perfect reconstruction, underscoring the need for plaintext-dependent key scheduling in future chaos-based cipher designs.

2. Description of the Target Encryption Scheme

This section provides a concise description of the encryption scheme proposed by Jain et al. [1], which serves as the target of the cryptanalytic attack developed in subsequent sections. The overall architecture is illustrated in Fig. 1, and the full encryption-decryption cycle is detailed below.

2.1. Chaotic Maps

Two distinct chaotic maps form the algorithmic backbone of the scheme. Arnold's Cat Map handles pixel-position shuffling during the permutation phase, while the Two-Dimensional Logistic Sine Coupling Map (2D-LSCM) drives keystream generation for the diffusion phase.

2.1.1. Two-Dimensional Logistic-Sine-Coupling Map (2D-LSCM)

The 2D-LSCM [32] is constructed by coupling the one-dimensional logistic map with the one-dimensional sine map. The logistic map is given by

$$x_{i+1} = r x_i (1 - x_i), \quad (1)$$

where r denotes the growth-rate parameter and $x_i \in (0, 1)$. The sine map takes the form

$$x_{i+1} = \alpha \sin(\pi x_i), \quad (2)$$

with control parameter $\alpha \in [0, 1]$. Coupling these two one-dimensional systems through a sine nonlinearity produces the 2D-LSCM:

$$\begin{cases} x_{i+1} = \sin(\pi(4\theta x_i(1 - x_i) + (1 - \theta) \sin(\pi y_i))), \\ y_{i+1} = \sin(\pi(4\theta y_i(1 - y_i) + (1 - \theta) \sin(\pi x_{i+1}))), \end{cases} \quad (3)$$

where $\theta \in [0, 1]$ is the coupling control parameter. By operating in a substantially higher-dimensional phase space than either constituent map, the 2D-LSCM achieves greater ergodicity and a wider chaotic parameter range [1].

2.1.2. Arnold's Cat Map

Arnold's Cat Map is a classical area-preserving automorphism of the torus, which, when applied to a square image, performs a deterministic re-ordering of pixel coordinates. For an image of size $n \times n$, the transformation is

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \bmod n, \quad (4)$$

with (x, y) and (x', y') denoting the pre-image and image pixel coordinates, respectively. The mapping is periodic with period $3n + 1$, meaning that repeated application returns the image to its original state after exactly $3n + 1$ iterations. The number of iterations P_i , satisfying $1 < P_i < 3n + 1$, constitutes the secret parameter K_1 for this component.

2.2. Key Generation

The complete secret key consists of two components. The first component, $K_1 = P_i$, specifies the iteration count for Arnold's Cat Map. The second component initializes the 2D-LSCM:

$$K_2 = \{x_0, y_0, r_{cp}, p_1, p_2\}, \quad (5)$$

where (x_0, y_0) are the map's initial values, r_{cp} is the coupling control parameter, and p_1, p_2 are auxiliary scalars used to derive two round-specific control parameters:

$$r_{cp1} = (r_{cp} \times p_1) \bmod 1, \quad r_{cp2} = (r_{cp} \times p_2) \bmod 1. \quad (6)$$

The combined length of K_2 amounts to 206 bits, comfortably exceeding the 100-bit threshold commonly cited as the minimum for brute-force resistance. Together, K_1 and K_2 constitute the full secret key of the scheme.

2.3. Encryption Process

Let P be a plain image of dimensions $M \times N$. Encryption is organized as two consecutive rounds, each comprising a permutation phase followed by a diffusion phase.

2.3.1. Permutation

The permutation phase introduces confusion by relocating pixel positions under the joint action of both chaotic maps.

1. Arnold's Cat Map is applied to P for P_i iterations per Eq. (4), producing an intermediate shuffled image.
2. A chaotic matrix X of size $M \times N$ is constructed by iterating the 2D-LSCM (Eq. (3)) with parameters drawn from K_2 .
3. Column-wise sorting of X in ascending order yields an index matrix I whose entries record the permutation induced by the sort.
4. For each row i , pixel values are gathered from positions $(I_{i,1}, 1), (I_{i,2}, 2), \dots, (I_{i,N}, N)$ of the Arnold-shuffled image.
5. A secondary index vector e , derived from sorting the chaotic values at the same positions, is used to re-order the gathered pixels, yielding the permuted image T .

The permuted image T depends exclusively on X and P_i ; no pixel value from P influences the permutation itself.

2.3.2. Diffusion

The diffusion phase modifies pixel values so that any single-pixel change in the plaintext propagates throughout the ciphertext. Let T be the permuted image and X the chaotic matrix. For an image sequence of length O (either M columns or N rows), the encrypted pixel E_i is

$$E_i = \begin{cases} P_1 + P_O + P_{O-1} + \lfloor X_i \times 2^{32} \rfloor \bmod F, & \text{if } i = 1, \\ P_2 + E_1 + P_O + \lfloor X_i \times 2^{32} \rfloor \bmod F, & \text{if } i = 2, \\ T_i + E_{i-1} + E_{i-2} + \lfloor X_i \times 2^{32} \rfloor \bmod F, & \text{if } i \in [3, O], \end{cases} \quad (7)$$

where $F = 255$ for 8-bit imagery and X_i is the i -th element of the chaotic matrix. The recursive dependence on E_{i-1} and E_{i-2} establishes an inter-pixel dependency chain that ensures downstream propagation of any modification. Column-wise and row-wise diffusion are applied in sequence. The entire permutation-diffusion cycle is then repeated using the second control parameter r_{cp_2} to yield the final cipher image C .

2.4. Decryption Process

Decryption constitutes the precise inverse of the encryption process, executed in reverse order. Given cipher image C and the key pair (K_1, K_2) , each diffusion stage is inverted first, followed by inversion of the corresponding permutation stage, for both rounds.

Inverse diffusion retrieves the permuted pixel values via

$$T_i = \begin{cases} E_1 - P_O - P_{O-1} - \lfloor X_i \times 2^{32} \rfloor \bmod F, & \text{if } i = 1, \\ E_2 - E_1 - P_O - \lfloor X_i \times 2^{32} \rfloor \bmod F, & \text{if } i = 2, \\ E_i - E_{i-1} - E_{i-2} - \lfloor X_i \times 2^{32} \rfloor \bmod F, & \text{if } i \in [3, O]. \end{cases} \quad (8)$$

Inverse permutation restores the original pixel arrangement by reversing both the 2D-LSCM sort and the Arnold's Cat Map steps. Since the Arnold map has period $3n + 1$, applying it a further $(3n + 1 - P_i)$ times achieves the inverse. The chaotic index matrices are regenerated from K_2 identically to the encryption procedure, and the sorting permutations are inverted. Processing the second round before the first recovers the original plaintext $\hat{P}$.

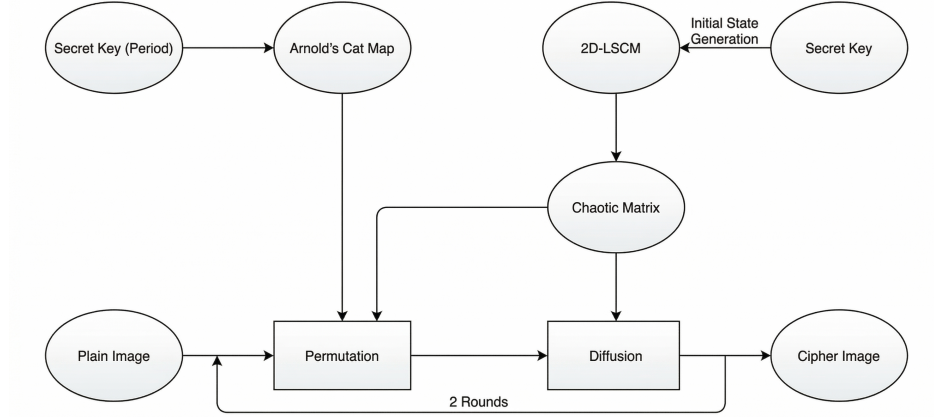


Figure 1: Block diagram of the encryption process of the target scheme [1]. The plain image undergoes two successive rounds of permutation (Arnold’s Cat Map + 2D-LSCM sorting) and diffusion (Eq. (7)) to produce the cipher image; decryption applies the inverse operations in reverse order.

3. Cryptanalysis

We now present a complete CPA against the scheme of Jain et al. [1], demonstrating that the two-round permutation-diffusion structure provides no actual resistance to a chosen-plaintext adversary. Both the equivalent keystream and the full permutation mapping can be extracted with a small oracle query budget, after which any ciphertext may be decrypted without the secret key.

The attack proceeds through three phases: (i) keystream extraction by encrypting a zero-valued image, (ii) permutation map reconstruction via structured probe images, and (iii) full decryption of any target ciphertext from the recovered parameters. The data complexity is $n + 1$ oracle queries for an $n \times n$ image, and the total computational cost is $\mathcal{O}(n^3)$, dominated by the Arnold map applications in Phase 2.

3.1. Adversarial Model and Attack Assumptions

The adversary is assumed to have black-box access to an encryption oracle $\mathcal{E}(\cdot)$ that, given any $n \times n$ image P , returns the corresponding ciphertext $C = \mathcal{E}(P)$ under an unknown fixed key (K_1, K_2) . The adversary has no knowledge of any internal parameters: x_0 , y_0 , θ , P_i , the chaotic matrix X , the chaotic sequence $\{x_i\}$, or the permutation map $\mathcal{M}$.

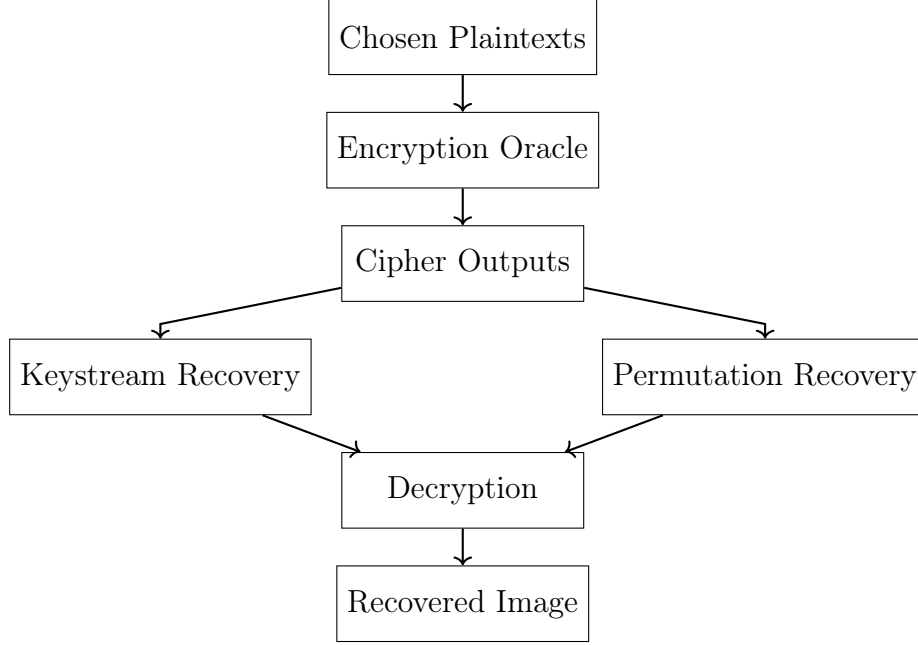


Figure 2: Block diagram of the proposed chosen-plaintext attack showing keystream recovery, permutation recovery, and final decryption without the secret key.

The exploitable structural weakness is the *plaintext-independence* of all key material. Because the 2D-LSCM sequence $\{x_i\}$ and the derived permutation map $\mathcal{M}$ are functions solely of the secret key—not of the image being encrypted—they remain identical across all images of the same dimensions under the same key. This renders them static equivalent keys, a well-documented vulnerability class in chaos-based cipher design [25].

3.2. Phase 1: Keystream Recovery via a Zero Plaintext

This phase exploits the encryption of an all-zero image to decouple the diffusion layer from the permutation layer and directly expose the keystream.

3.2.1. Structural Observation

Let $\mathbf{0}$ denote the $n \times n$ all-zero image and let $C^{(0)} = \mathcal{E}(\mathbf{0})$.

Step 1 (Arnold’s Cat Map acts as the identity on a zero image). Arnold’s Cat Map (Eq. (4)) permutes pixel *coordinates* while leaving pixel *values* unchanged. Because every pixel of $\mathbf{0}$ carries the value zero, the map

produces the same all-zero image irrespective of the iteration count:

$$\text{ACM}^{P_i}(\mathbf{0}) = \mathbf{0}. \quad (9)$$

Step 2 (2D-LSCM column permutation acts as the identity on a zero image). The column-wise permutation step rearranges rows within column j according to a permutation σ_j derived from chaotic matrix X :

$$T[:, j] = \text{ACM}^{P_i}(P)[\sigma_j, j], \quad j = 1, \dots, n. \quad (10)$$

When $P = \mathbf{0}$, every element selected by the permutation equals zero, so:

$$T^{(0)} = \mathbf{0}. \quad (11)$$

Step 3 (Diffusion collapses to a pure keystream recurrence). With $T_i^{(0)} = 0$ for all i , Eq. (7) reduces to

$$C_i^{(0)} = \begin{cases} X_i, & i = 1, \\ C_1^{(0)} + X_i \bmod F, & i = 2, \\ C_{i-1}^{(0)} + C_{i-2}^{(0)} + X_i \bmod F, & i \geq 3, \end{cases} \quad (12)$$

which expands to

$$\begin{aligned} C_1^{(0)} &= X_1, \\ C_2^{(0)} &= X_1 + X_2 \bmod F, \\ C_i^{(0)} &= C_{i-1}^{(0)} + C_{i-2}^{(0)} + X_i \bmod F, \quad i \geq 3. \end{aligned} \quad (13)$$

Rearranging yields the keystream recovery formula:

$$\hat{X}_i = \begin{cases} C_1^{(0)}, & i = 1, \\ (C_2^{(0)} - C_1^{(0)}) \bmod F, & i = 2, \\ (C_i^{(0)} - C_{i-1}^{(0)} - C_{i-2}^{(0)}) \bmod F, & i \geq 3. \end{cases} \quad (14)$$

3.2.2. Correctness of Keystream Recovery

Every value $\hat{X}_i$ recovered by Eq. (14) satisfies $\hat{X}_i = X_i$ for $i = 1, \dots, N$.

Case $i = 1$:

From Eq. (13),

$$C_1^{(0)} = X_1 \bmod F$$

Hence,

$$\hat{X}_1 = C_1^{(0)} = X_1.$$

Case $i = 2$:

$$C_2^{(0)} = (C_1^{(0)} + X_2) \bmod F$$

Thus,

$$\hat{X}_2 = (C_2^{(0)} - C_1^{(0)}) \bmod F = X_2.$$

Case $i \geq 3$:

Assume $e_k = X_k$ for all $k < i$. Then,

$$C_i^{(0)} = (C_{i-1}^{(0)} + C_{i-2}^{(0)} + X_i) \bmod F$$

Hence,

$$\hat{X}_i = (C_i^{(0)} - C_{i-1}^{(0)} - C_{i-2}^{(0)}) \bmod F = X_i.$$

The complete keystream is thus recovered exactly from a single oracle query.

3.3. Phase 2: Permutation Map Reconstruction

With the keystream $\{\hat{X}_i\}$ in hand, we show how to reconstruct the full permutation map $\mathcal{M}$ using exactly n additional oracle queries.

3.3.1. Permutation Structure

From Section 2.3, the permutation stage applies Arnold's Cat Map for P_i iterations and then performs a column-wise sorting step driven by the chaotic matrix X . Specifically, for column j , if $\sigma_j = \text{argsort}(X[:, j])$ is the permutation induced by sorting that column, the output satisfies

$$T[:, j] = S[\sigma_j, j], \quad (15)$$

where $S = \text{ACM}^{P_i}(P)$. The attacker seeks to determine $\mathcal{M}[:, j] = \sigma_j$ for every column.

3.3.2. Probe Plaintext Design

For each column index $j \in \{1, \dots, n\}$, a probe image $P^{(j)}$ is defined as

$$P_{r,c}^{(j)} = \begin{cases} r, & c = j, \\ 0, & c \neq j, \end{cases} \quad (16)$$

so that column j encodes the row index sequence $0, 1, \dots, n-1$ and every other column is identically zero.

To neutralize the Arnold preprocessing, the probe submitted to the oracle is the P_i -step pre-image of $P^{(j)}$ under the Arnold map:

$$Q^{(j)} = \text{ACM}^{-P_i}(P^{(j)}) = \text{ACM}^{(3n+1-P_i)}(P^{(j)}), \quad (17)$$

exploiting the periodicity of Arnold's Cat Map to compute the inverse via forward iteration.

3.3.3. Oracle Output and Permutation Recovery

When the oracle processes $Q^{(j)}$, the Arnold step undoes the pre-imaging:

$$S^{(j)} = \text{ACM}^{P_i}(Q^{(j)}) = \text{ACM}^{P_i}(\text{ACM}^{-P_i}(P^{(j)})) = P^{(j)}. \quad (18)$$

Applying the 2D-LSCM column sort to $S^{(j)} = P^{(j)}$:

$$T^{(j)}[:, k] = P^{(j)}[\sigma_k, k], \quad k = 1, \dots, n. \quad (19)$$

Examining column j and using $P_{r,j}^{(j)} = r$:

$$T^{(j)}[\ell, j] = P^{(j)}[\sigma_j(\ell), j] = \sigma_j(\ell), \quad \ell = 1, \dots, n. \quad (20)$$

All remaining columns $k \neq j$ satisfy $T^{(j)}[:, k] = \mathbf{0}$ since $P^{(j)}[:, k] = \mathbf{0}$. Consequently, column j of the intermediate image carries precisely the permutation indices σ_j .

The oracle then applies diffusion to yield $C^{(j)} = \mathcal{E}(Q^{(j)})$. Inverting the diffusion using $\hat{X}$:

$$\hat{T}_i^{(j)} = \begin{cases} (C_1^{(j)} - C_N^{(j)} - C_{N-1}^{(j)} - \hat{X}_1) \bmod F, & i = 1, \\ (C_2^{(j)} - C_1^{(j)} - C_N^{(j)} - \hat{X}_2) \bmod F, & i = 2, \\ (C_i^{(j)} - C_{i-1}^{(j)} - C_{i-2}^{(j)} - \hat{X}_i) \bmod F, & i \geq 3, \end{cases} \quad (21)$$

from which the permutation indices for column j are read directly:

$$\hat{\sigma}_j(\ell) = \hat{T}^{(j)}[\ell, j], \quad \ell = 1, \dots, n. \quad (22)$$

Iterating over all $j = 1, \dots, n$ completes reconstruction of $\hat{\mathcal{M}}$.

The recovered map satisfies $\hat{\mathcal{M}} = \mathcal{M}$ exactly.

Eq. (20) establishes $T^{(j)}[\ell, j] = \sigma_j(\ell)$. Since Proposition 1 guarantees $\hat{X}_i = X_i$ for all i , the inverse diffusion in Eq. (21) recovers $\hat{T}^{(j)} = T^{(j)}$ without error, whence $\hat{\sigma}_j = \sigma_j$ for every j .

3.3.4. Handling the Final Two Boundary Terms

The first two positions of the diffusion recurrence (Eq. (7)) depend on T_O and T_{O-1} —the last two elements of the permuted flat sequence—rather than on prior encrypted pixels. For the probe images $Q^{(j)}$, these boundary values are zero because the flattened permuted sequence has a non-zero single column, and the trailing entries happen to be zero under row-major ordering. Accordingly, Eq. (21) applies unchanged. When decrypting an arbitrary target ciphertext, these boundary terms are correctly handled by iterating the inverse diffusion from index $i = N$ down to $i = 1$.

3.4. Phase 3: Complete Decryption of Arbitrary Ciphertexts

Armed with the recovered keystream $\hat{X}$ and permutation map $\hat{\mathcal{M}}$, decryption of any ciphertext $C = \mathcal{E}(P)$ proceeds in three steps without reference to the secret key.

Step 1 (Inverse diffusion). Compute the permuted image $\hat{T}$ from C via the inverse diffusion operator:

$$\hat{T} = \mathcal{D}_{\hat{X}}(C), \quad (23)$$

where $\mathcal{D}_{\hat{X}}$ denotes Eq. (8) instantiated with $\hat{X}$. By Proposition 1, $\hat{T} = T$ exactly.

Step 2 (Inverse column permutation). For each column j , reverse the 2D-LSCM sort using $\hat{\mathcal{M}}$:

$$\hat{S}[\hat{\sigma}_j(\ell), j] = \hat{T}[\ell, j], \quad \ell = 1, \dots, n, \quad (24)$$

recovering $\hat{S} = S = \text{ACM}^{P_i}(P)$.

Step 3 (Inverse Arnold's Cat Map). Apply the Arnold map $(3n + 1 - P_i)$ times to $\hat{S}$:

$$\hat{P} = \text{ACM}^{-P_i}(\hat{S}) = \text{ACM}^{(3n+1-P_i)}(\hat{S}). \quad (25)$$

Since $\hat{S} = \text{ACM}^{P_i}(P)$ and Arnold's Cat Map is invertible, $\hat{P} = P$ exactly.

Remark on P_i in Algorithm 1. Phase 2 requires knowledge of P_i to construct the probe pre-images $Q^{(j)}$. If P_i is not known, it can be identified by a linear scan: for each candidate $\tilde{P}_i \in \{1, \dots, 3n\}$, the attacker submits one structured probe and verifies whether column j of the recovered intermediate image matches the expected pattern $\{0, 1, \dots, n - 1\}$. The correct value is

found in at most $3n - 1$ additional queries, which does not alter the overall polynomial complexity of the attack.

Algorithm 1 Chosen-Plaintext Attack on the Target Scheme

Input: Encryption oracle $\mathcal{E}(\cdot)$; target ciphertext C ; image size n .

Output: Recovered plaintext $\hat{P}$.

```

// Phase 1: Keystream Recovery
1  $C^{(0)} \leftarrow \mathcal{E}(\mathbf{0}_{n \times n})$  Compute  $\hat{X}_i$  for all  $i$  using Eq. (14)

// Phase 2: Permutation Map Recovery
2 for  $j = 1$  to  $n$  do
3   Construct  $P^{(j)}$  Set  $P_{r,j}^{(j)} \leftarrow r$  for all  $r$  Set  $P_{r,k}^{(j)} \leftarrow 0$  for all  $k \neq j$ 
4   Compute  $Q^{(j)} \leftarrow \text{ACM}^{(3n+1-P_i)}(P^{(j)})$ 
5    $C^{(j)} \leftarrow \mathcal{E}(Q^{(j)})$ 
6    $\hat{T}^{(j)} \leftarrow \mathcal{D}_{\hat{X}}(C^{(j)})$ 
7    $\hat{\sigma}_j \leftarrow \hat{T}^{(j)}[:, j]$ 
8 end

// Phase 3: Decryption of Target Ciphertext
9  $\hat{T} \leftarrow \mathcal{D}_{\hat{X}}(C)$  for  $j = 1$  to  $n$  do
10   $\hat{S}[\hat{\sigma}_j(:, j), j] \leftarrow \hat{T}[:, j]$ 
11 end
12  $\hat{P} \leftarrow \text{ACM}^{(3n+1-\hat{P}_i)}(\hat{S})$  return  $\hat{P}$ 

```

3.5. Complexity Analysis

Data complexity. The entire attack consumes $1 + n$ oracle queries: one zero-image query in Phase 1 and one query per image column in Phase 2. For a 256×256 image this amounts to 257 queries—negligible from a practical standpoint.

Computational complexity. Phase 1 inverts the diffusion over all $N = n^2$ pixels in $\mathcal{O}(n^2)$ time. Phase 2 constructs n probe pre-images, each requiring up to $3n$ Arnold map iterations on an $n \times n$ image; since each iteration costs $\mathcal{O}(n^2)$, Phase 2 runs in $\mathcal{O}(n^3)$ in total. Phase 3 requires $\mathcal{O}(n^2)$. The dominant cost is therefore $\mathcal{O}(n^3)$, matching the complexity of the encryption algorithm itself.

Memory complexity. The attacker need only store the recovered keystream $\hat{X}$ (n^2 entries) and the permutation map $\hat{\mathcal{M}}$ (n^2 entries), for a total of $\mathcal{O}(n^2)$

space.

Table 1: Attack complexity compared to the encryption cost.

	Encryption	CPA Attack
Oracle queries	—	$n + 1$
Arithmetic operations	$\mathcal{O}(n^3)$	$\mathcal{O}(n^3)$
Memory	$\mathcal{O}(n^2)$	$\mathcal{O}(n^2)$

4. Experimental Simulation of the Chosen-Plaintext Attack

This section presents an end-to-end experimental evaluation of the proposed CPA against the scheme of Jain et al. [1]. All experiments were conducted in Python 3 on a 256×256 grayscale medical image (the standard Cameraman benchmark). Strict CPA conditions were enforced throughout: the attacker was permitted access only to the encryption oracle $\mathcal{E}(\cdot)$ and possessed no information whatsoever about the secret key.

4.1. Experimental Setup

The target image was a 256×256 grayscale frame with 8-bit pixel values in $[0, 255]$, comprising $N = 65,536$ pixels in total. The oracle was configured with the following fixed key, withheld from the attacker:

$$K_1 = P_i = 10, \quad K_2 = \{x_0 = 0.8, y_0 = 0.5, \theta = 0.99\}. \quad (26)$$

These parameters were used solely to instantiate the oracle; the attack procedure at no point queries or assumes any knowledge of these values. The cipher image $C = \mathcal{E}(P)$ generated under Eq. (26) bears no visual resemblance to the plaintext, consistent with the scheme’s stated confusion and diffusion properties.

4.2. Phase 1: Keystream Recovery

The all-zero image $\mathbf{0}_{256 \times 256}$ was submitted to the oracle, yielding $C^{(0)} = \mathcal{E}(\mathbf{0})$. The equivalent keystream $\hat{X}$ was then computed from $C^{(0)}$ via Eq. (14), with no knowledge of K_2 required.

4.2.1. Chaotic Diffusion Sequence

The 2D-LSCM sequence $\{x_i\}$ generated internally by the oracle (unknown to the attacker) had the following first 50 values:

$$\{x_i\}_{i=1}^{50} = \{0.3581, 0.2567, 0.6761, 0.3763, 0.2196, \\ 0.8440, 0.9952, 0.0681, 0.7209, 0.5955, \\ 0.1320, 0.9896, 0.1407, 0.9979, 0.0375, \\ 0.4384, 0.0494, 0.5673, 0.0768, 0.7750, \\ 0.8083, 0.9352, 0.7058, 0.5164, 0.0328, \\ 0.4044, 0.1218, 0.9774, 0.2970, 0.4940, \dots\}$$

The corresponding keystream values

$$X_i = \lfloor |x_i| \cdot 2^{32} \rfloor \bmod 256$$

(the values actually embedded in the diffusion) coincide exactly with the attacker's recovered values $\hat{X}_i$. The first 50 comparison pairs are:

$$\{(X_i, \hat{X}_i)\}_{i=1}^{50} = \{(17, 17), (13, 13), (31, 31), (171, 171), (67, 67), \\ (208, 208), (8, 8), (255, 255), (98, 98), (238, 238), \\ (157, 157), (114, 114), (82, 82), (10, 10), (50, 50), \\ (20, 20), (86, 86), (152, 152), (252, 252), (227, 227), \\ (140, 140), (89, 89), (252, 252), (116, 116), (181, 181), \\ (153, 153), (130, 130), (42, 42), (253, 253), (81, 81), \\ (254, 254), (205, 205), (173, 173), (168, 168), (243, 243), \\ (44, 44), (165, 165), (135, 135), (174, 174), (191, 191), \\ (14, 14), (42, 42), (66, 66), (220, 220), (40, 40), \\ (25, 25), (134, 134), (206, 206), (68, 68), (215, 215)\}.$$

4.2.2. Phase 1 Result Summary

All $N = 65,536$ keystream values were recovered with perfect accuracy from a single oracle call, confirming Proposition 1.

Table 2: Keystream recovery metrics after one oracle query.

Metric	Value
Oracle queries used	1
Total keystream values recovered	65,536
Perfect match ($\hat{X} = X$)	True
Max absolute deviation	0
Mean absolute deviation	0.000000

4.3. Phase 2: Permutation Map Reconstruction

For each column $j \in \{1, \dots, 256\}$, the attacker assembled the structured probe $P^{(j)}$ (Eq. (16)), computed its Arnold pre-image $Q^{(j)}$ (Eq. (17)), and submitted $Q^{(j)}$ to the oracle. Each returned ciphertext was inverse-diffused with $\hat{X}$ to read off the permutation indices $\hat{\sigma}_j$ for the corresponding column. This phase consumed exactly 256 oracle queries.

4.3.1. Permutation Map Comparison

Table 4 displays the upper-left 8×8 submatrix of both the true permutation map $\mathcal{M}$ (inaccessible to the attacker) and the recovered map $\hat{\mathcal{M}}$. Every entry is identical.

4.3.2. Phase 2 Result Summary

Table 3: Permutation map recovery metrics after 256 oracle queries.

Metric	Value
Oracle queries used	256
Columns recovered	256 / 256
Perfect match ($\hat{\mathcal{M}} = \mathcal{M}$)	True
Max absolute deviation	0
Mean absolute deviation	0.000000

All 256 columns of the permutation map were recovered without error, confirming Proposition 2.

Table 4: Top-left 8×8 block of the true permutation map $\mathcal{M}$ (top) and the recovered permutation map $\hat{\mathcal{M}}$ (bottom). All values match exactly.

True permutation map $\mathcal{M}$ (first 8×8)							
59	185	57	186	253	182	113	46
83	6	67	26	23	24	197	181
49	135	29	128	176	172	53	20
168	132	81	211	222	194	55	75
175	90	130	58	177	163	170	93
77	49	216	29	208	101	168	243
41	16	17	83	1	135	128	10
148	64	251	69	61	188	80	109
Recovered permutation map $\hat{\mathcal{M}}$ (first 8×8)							
59	185	57	186	253	182	113	46
83	6	67	26	23	24	197	181
49	135	29	128	176	172	53	20
168	132	81	211	222	194	55	75
175	90	130	58	177	163	170	93
77	49	216	29	208	101	168	243
41	16	17	83	1	135	128	10
148	64	251	69	61	188	80	109

4.4. Phase 3: Full Decryption Without the Secret Key

Using $(\hat{X}, \hat{\mathcal{M}})$, the attacker executed the three-step decryption procedure of Section 3.4: (i) inverse diffusion of C via $\hat{X}$ (Eq. (23)); (ii) inverse column permutation via $\hat{\mathcal{M}}$ (Eq. (24)); (iii) inverse Arnold map application (Eq. (25)). At no point was any component of (K_1, K_2) consulted.

4.4.1. Visual and Quantitative Results

Fig. 3 presents a four-panel comparison covering the original plaintext, the target ciphertext, the zero-image ciphertext from Phase 1, and the CPA-recovered image.

Pixel-level fidelity is measured by

$$\text{MaxDiff} = \max_{i,j} |P_{ij} - \hat{P}_{ij}| = 0, \quad \text{MeanDiff} = \frac{1}{N} \sum_{i,j} |P_{ij} - \hat{P}_{ij}| = 0.000000, \quad (27)$$

establishing that $\hat{P} = P$ exactly, i.e., reconstruction is pixel-perfect.

4.4.2. Phase 3 Result Summary

Table 5: CPA decryption accuracy metrics.

Metric	Value
Oracle queries in total ($1 + 256$)	257
CPA-decrypted image matches original ($\hat{P} = P$)	True
Max pixel difference	0
Mean pixel difference	0.000000

4.5. Overall Attack Summary

Table 6 summarizes data complexity, oracle queries, and recovery accuracy across all three phases.

Table 6: Summary of the chosen-plaintext attack on the 256×256 Cameraman test image. Total oracle queries: $n + 1 = 257$.

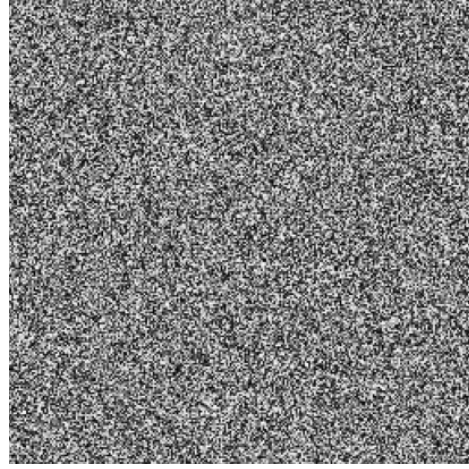
Phase	Goal	Oracle calls	Items recovered	Perfect match
1	Keystream $\hat{X}$	1	65,536 values	Yes ($\Delta = 0$)
2	Perm. map $\hat{\mathcal{M}}$	256	256×256 indices	Yes ($\Delta = 0$)
3	Plaintext $\hat{P}$	0	256×256 pixels	Yes ($\Delta = 0$)
Total		257		Exact recovery

The results confirm that any plaintext encrypted under the target scheme can be recovered completely and without loss using only $n + 1 = 257$ oracle queries on a 256×256 image. The scheme is therefore *completely insecure* within the CPA model, notwithstanding its 206-bit key space. Satisfactory performance on NPCR, UACI, and entropy measures is *not sufficient* to guarantee cryptographic security—a well-established principle [25] that this case study vividly illustrates.

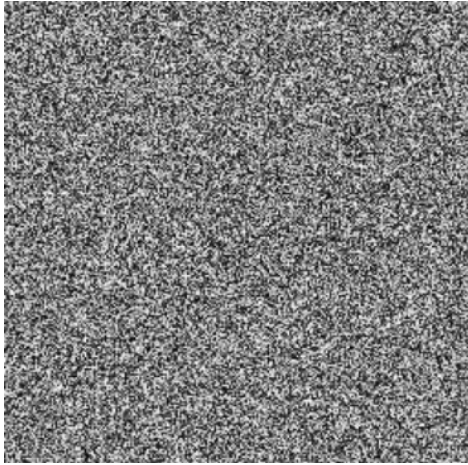
The implementation used for all experiments is available at Colab Notebook.



(a) Original image P



(b) Encrypted $C = \mathcal{E}(P)$



(c) Blank cipher $C^{(0)}$



(d) CPA-decrypt $\hat{P}$

Figure 3: Full chosen-plaintext attack pipeline. (a) Original plaintext image. (b) Encrypted ciphertext submitted to oracle. (c) Zero-image ciphertext used in Phase 1 to extract the keystream. (d) Plaintext $\hat{P}$ recovered by the attacker *without any access to the secret key*, matching (a) with zero pixel error.

4.6. Root Cause Analysis: Why the Scheme is Broken

Three mutually reinforcing design defects collectively explain the attack's success.

Defect 1: Keystream that does not depend on the plaintext. The 2D-LSCM sequence $\{x_i\}$ is entirely determined by $K_2 = \{x_0, y_0, r_{cp}, p_1, p_2\}$ with no contribution from the plaintext image. Consequently, the embedded keystream $X = \lfloor |x_i| \cdot 2^{32} \rfloor \bmod F$ takes the same values for every image encrypted under an identical key, regardless of content. A cipher whose keystream is independent of the plaintext is functionally a reusable pad and is unconditionally vulnerable to chosen-plaintext analysis [25].

Defect 2: Fully decoupled permutation and diffusion stages. Because permutation precedes diffusion with no feedback mechanism linking one stage’s output to the other’s key material, each stage is individually invertible. This is the canonical weakness of uncoupled permutation-diffusion architectures [16]. Had the keystream been computed as a function of the post-permutation image rather than the pre-permutation plaintext, encrypting a zero image would not have reduced the diffusion output to a transparent function of the keystream alone, and the zero-image attack would fail.

Defect 3: Publicly invertible Arnold preprocessing. Arnold’s Cat Map is a well-known, fully public, deterministic transformation parameterized only by the iteration count P_i . The adversary can compute both its forward map and its inverse freely. This ability is exploited to pre-cancel the Arnold layer in the oracle: by submitting the Arnold pre-image of each probe, the attacker causes the oracle’s internal Arnold application to restore the probe exactly, reducing the effective permutation to the 2D-LSCM sort alone, which is then directly revealed by the diffusion output.

Combined consequence. The co-occurrence of all three defects means that the scheme’s sole security asset is the secrecy of the key itself, with no algorithmic hardness contributing to protection. An adversary holding oracle access to the scheme can extract the complete cryptographic state—keystream X and permutation map $\mathcal{M}$ —in merely $n + 1$ queries, and subsequently decrypt any ciphertext in $\mathcal{O}(n^2)$ time.

5. Conclusion

In this paper, the chaotic image encryption scheme based on Arnold’s Cat Map and a Two-Dimensional Logistic Sine Coupling Map is analyzed in detail. Although the scheme employs chaotic maps and follows a permutation–diffusion structure, it is shown that it cannot withstand a chosen-plaintext attack due to several inherent design flaws.

Specifically, the existence of a plaintext-independent keystream allows the attacker to recover an equivalent key using a single chosen plaintext. Furthermore, since the permutation and diffusion stages are completely independent, each component can be separately inverted. Based on these observations, a practical chosen-plaintext attack is constructed. First, the equivalent keystream is extracted using an all-zero plaintext. Second, multiple probe images are used to recover the permutation mapping. Finally, the combination of these recovered components enables exact decryption of any ciphertext.

Both theoretical analysis and experimental results confirm that the proposed attack can fully break the encryption scheme with low data complexity. For a grayscale image of size 256×256 , the attack requires only $n + 1$ oracle queries and achieves perfect reconstruction without error, despite the scheme having a large nominal key space and passing standard statistical tests.

The cryptanalysis presented in this work highlights that passing statistical evaluations does not guarantee resistance against algebraic or chosen-plaintext attacks. To enhance security, future designs of chaos-based image encryption schemes should ensure that keystream generation depends on the plaintext and that permutation and diffusion processes are tightly coupled.

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Use of AI Tools. During the preparation of this work, the authors used ChatGPT (OpenAI) to improve readability, refine language, and assist in structuring portions of the manuscript. All generated content was carefully reviewed and edited by the authors. The authors take full responsibility for the accuracy, originality, and integrity of the final manuscript.

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